Branch-and-Bound Tensor Network for Combinatorial Optimization

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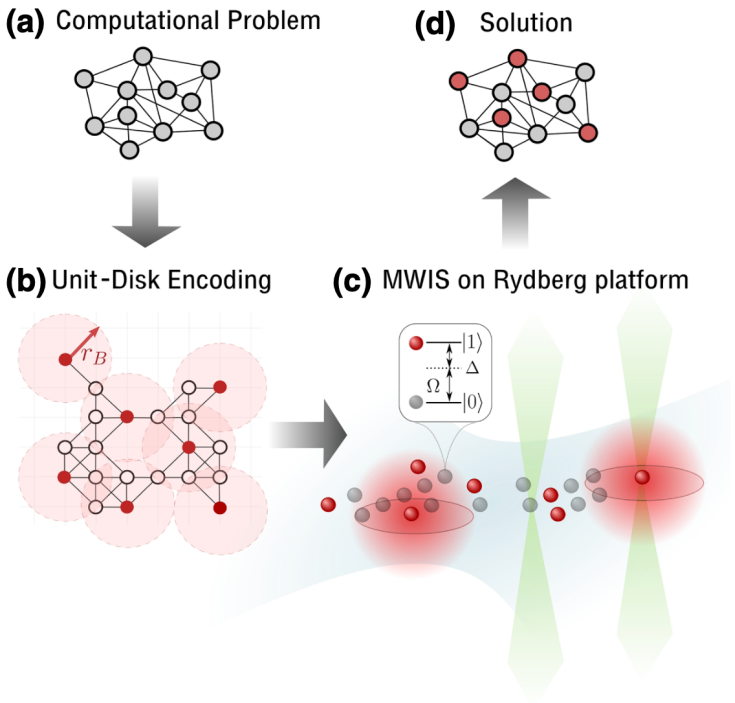
Motivation

- Combinatorial optimization problems such as maximum independent set and spin-glasses are important.
- Exact optimization (NP-hard) and counting of ground states (#P-hard) expose their solution-space structure.

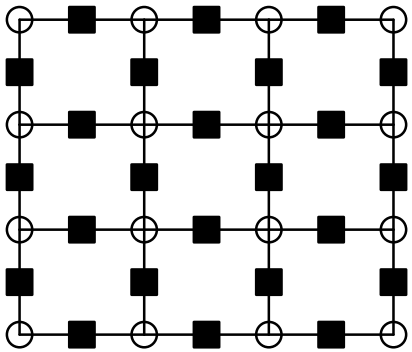


Maximum independent set and Rydberg atom arrays[1].

[1] *PRX Quantum 4, 010316 (2023).*

Tropical Tensor Networks and Slicing

- Tropical tensor networks[2] exploit GPU parallelism, but memory grows exponentially with treewidth.
- Slicing reduces memory by fixing variables, but each extra fixed variable doubles the number of subproblems, greatly increasing runtime.



Normal algebra → Partition function

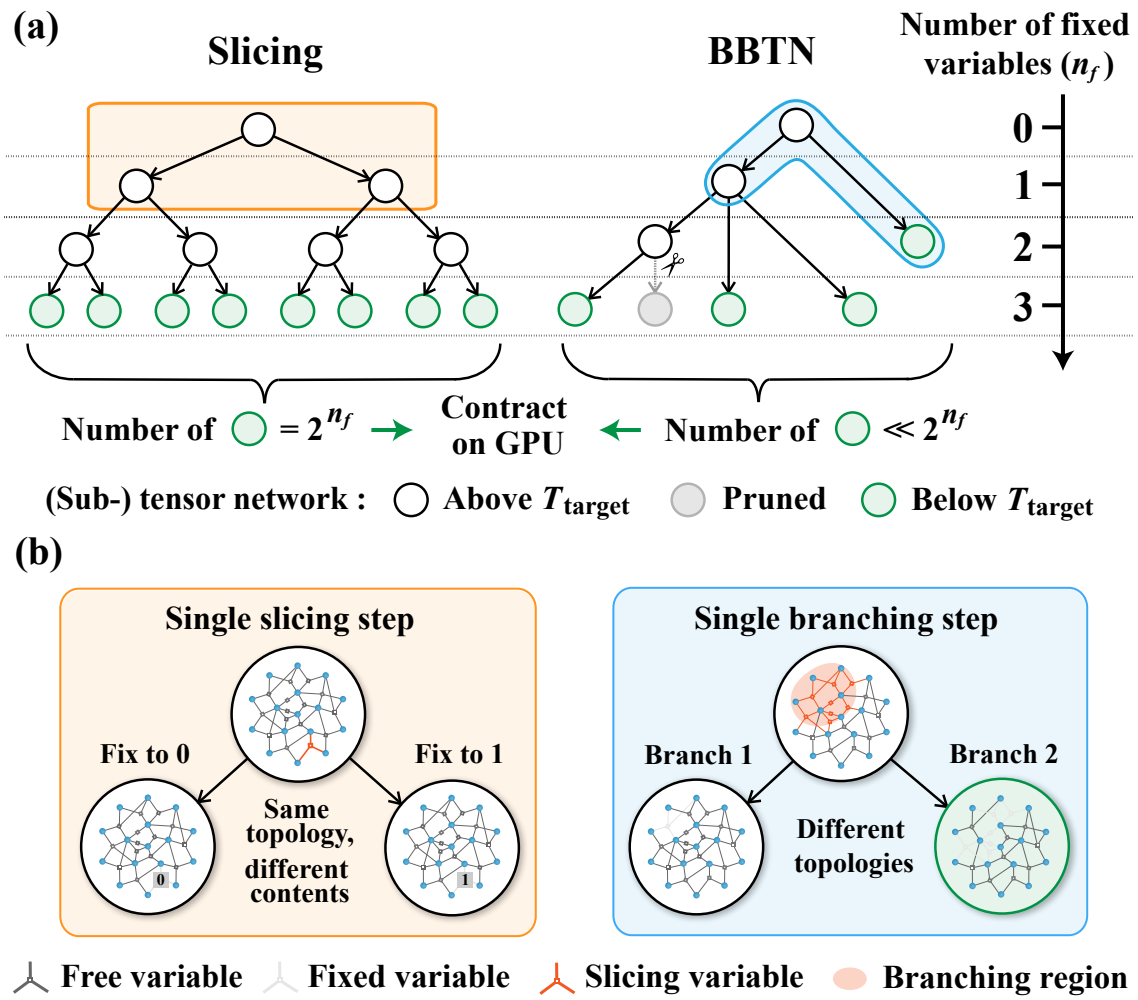
$$x \odot y = x + y$$
$$x \oplus y = \max(x, y)$$
→ Ground state

$$(x_1, n_1) \odot (x_2, n_2) = (x_1 + x_2,\ n_1 n_2)$$
$$(x_1, n_1) \oplus (x_2, n_2) = \begin{cases} (x_1,\ n_1 + n_2), & x_1 = x_2 \\ (x_1,\ n_1), & x_1 > x_2 \\ (x_2,\ n_2), & x_1 < x_2 \end{cases}$$
→ Ground-state degeneracy

[2] *Phys. Rev. Lett. 126, 090506 (2021).*

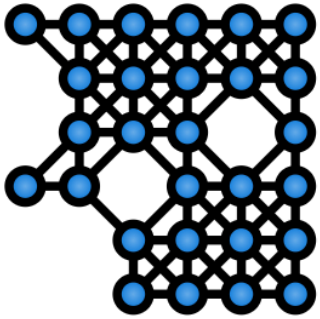
Brach-and-Bound Tensor Network

The main idea of BBTN is to reduce number of subproblems by using branch-and-bound search.

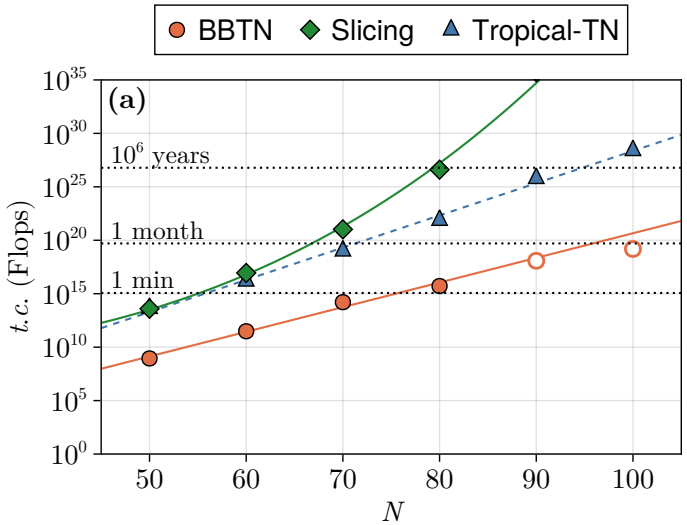


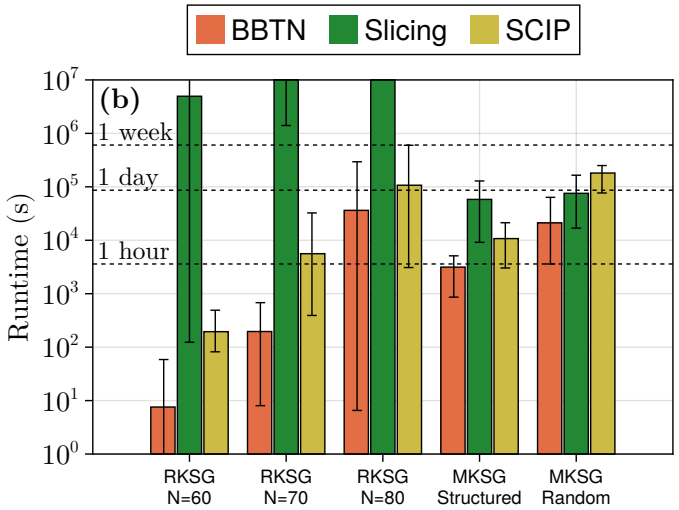
Ground State Optimization

- Exact maximum independent set on King's subgraphs ($\rho = 0.8$) up to 100×100.
- More than $30\times$ faster than leading integer-programming solvers on hard instances.



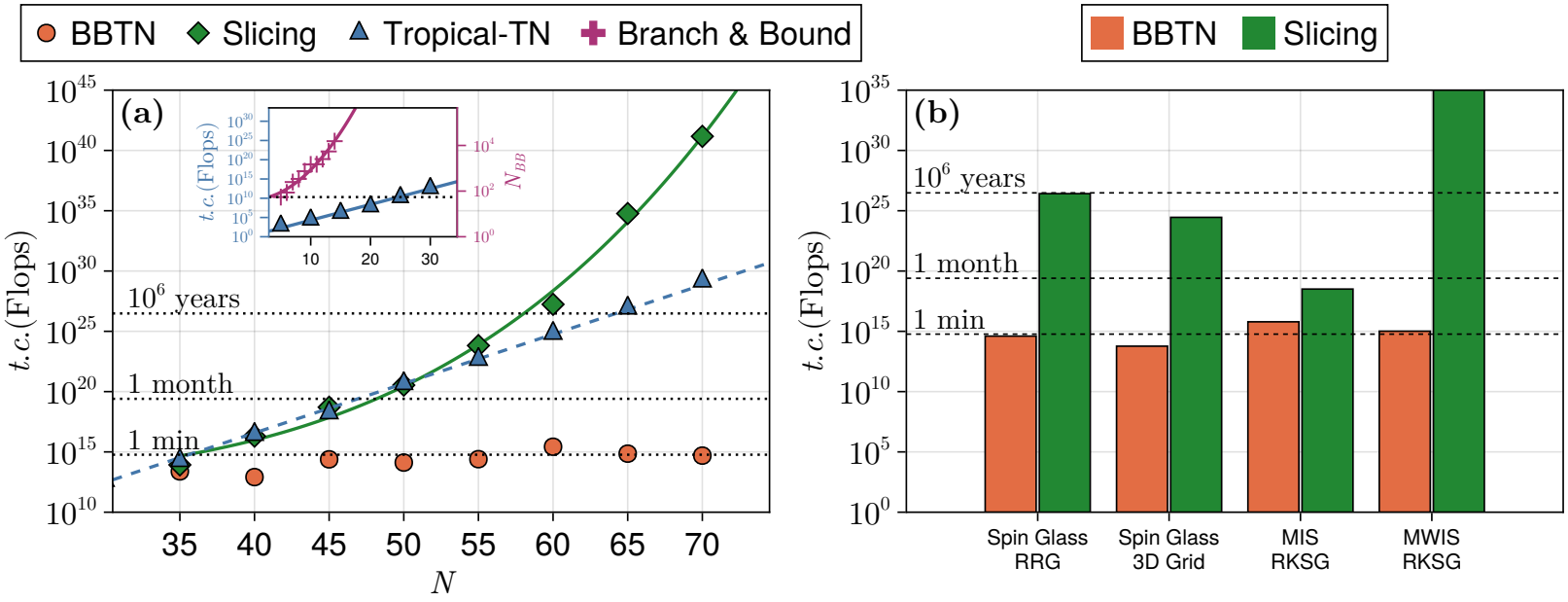
King's subgraph





Ground State Counting

- Exact ground-state degeneracy counting for $\pm J$ spin glasses up to 64×64.
- Vanilla branch-and-bound only reaches $N \leq 15$, while slicing is estimated to require more than 10^4 years at $N = 60$.



Take Home Message

- BBTN combines branch-and-bound search with tropical tensor-network contraction for exact ground-state characterization.
- This removes the main space-complexity bottleneck and pushes exact counting/optimization to much larger instances.
- In hard regimes, BBTN compresses computations from years to minutes and can outperform leading integer-programming solvers by more than 30×.

arxiv.org/abs/2602.05470

TensorBranching.jl

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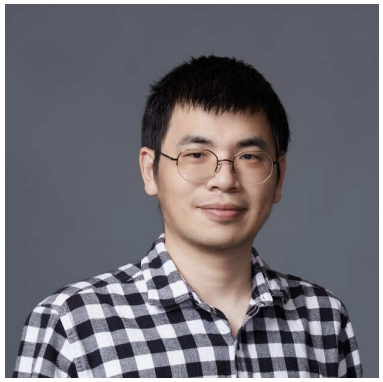

Prof. Jinguo Liu,
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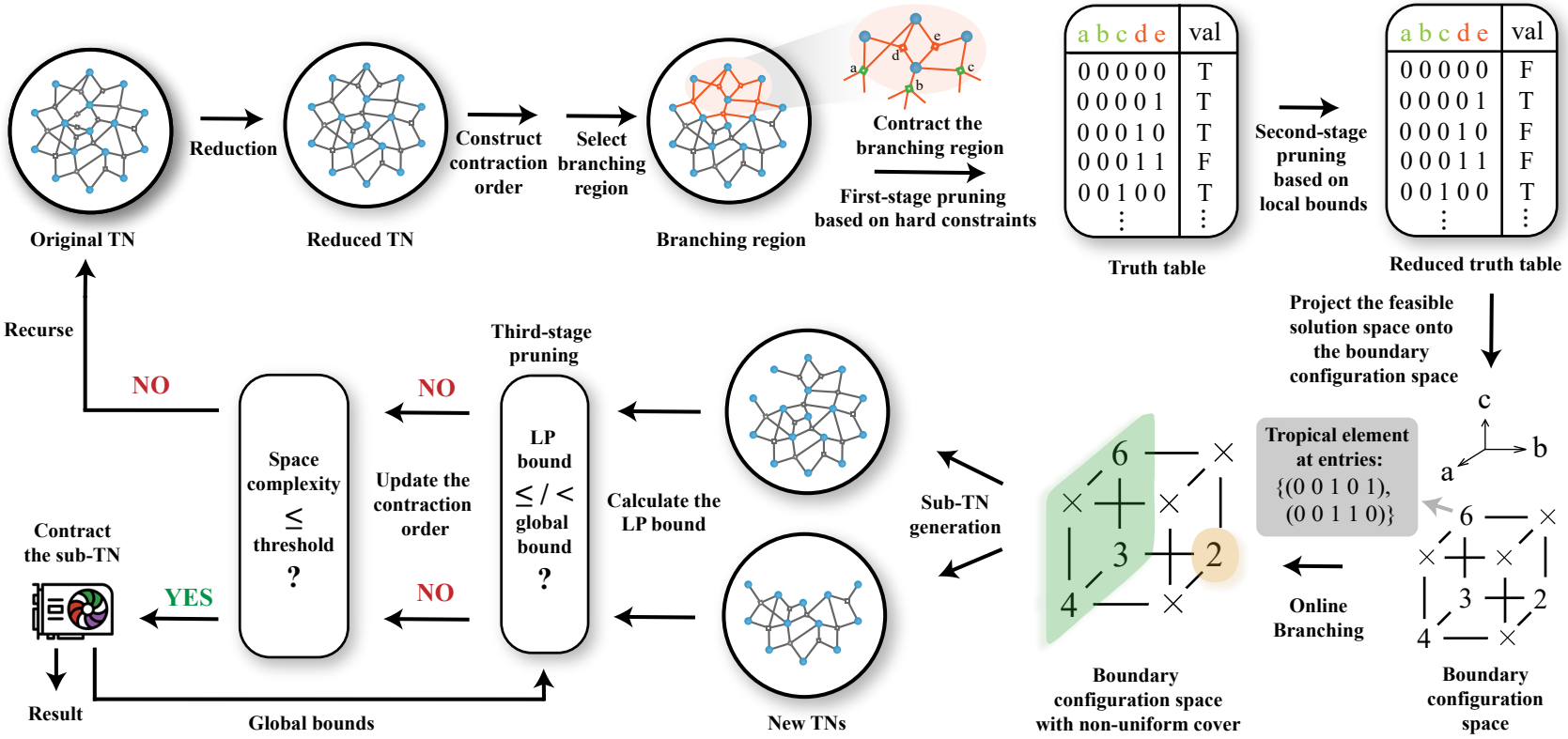
ITP, CAS

Thank you!

Brach-and-Bound Tensor Network: Algorithm



References

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PRX Quantum, 4(1):010316, 2023.